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Changing of the shape and structure of Cu nanoclusters generated from a gas phase: MD simulations

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ABSTRACT

The gas-phase condensation of 85,000 Cu atoms is examined by molecular dynamics simulation with a tight-binding potential. A detailed study of the evolution of the system cooled at a constant rate from 1000 K to 77 K is presented. The results are used to identify four distinct stages of the evolution from a hot atomic gas to a few synthesized particles. The effect of the subsequent thermal treatment on the shape and structure of synthesized particles was studied by simulating their gradual heating in a range of 100–1200 K. It is concluded that short-term heating leads to significant ordering of the internal structure in 70% of agglomerated nanoparticles with the predominant formation of spherical shapes. In order to explain this result, the main mechanisms of cluster formation from the gas phase have been analyzed and it is found that the agglomeration temperature plays the main role in the formation of clusters with unified shape and structure.

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1. Introduction

Studying individual nanoparticles and the nanostructured state is currently one of the most intensively developing fields of research in physics, chemistry, and engineering. Great scientific and practical interest in such research results from the unique properties of nanoparticles that are already or will be widely used in the immediate future for the fabrication of miniature electronic devices, and production of new materials.

Copper clusters hold a most unique position among the whole range of currently used metal nanoparticles. These particles have many unique properties and are relatively inexpensive for manufacturing. Relating to conducting properties copper nanosized particles may compete with silver. Also the range of their usage is wide in the form of catalysts (Dubey et al., 2012). However the formation of nanoparticles of copper with implementation of application area dependent properties specifies very strict requirements to their producing and processing. One of the promising techniques of nanodispersed particles synthesis process is metals evaporation and condensation in the inert gas atmosphere. Theoretically, this technique makes it possible to synthesize nanoparticles with controlled chemical composition (Starsich et al., 2014), defect density, structure, and size distribution. Moreover, it is relatively easy to control the parameters of gas-phase synthesis under experimental conditions (Kauffeldt & Kauffeldt, 2006; Weber et al., 2006; Seipenbusch et al., 2007).

However one of the main characteristics of vapor phase synthesis method is the substantial inner structure and external shape discontinuity of resulting nanoparticles. Therefore, depending on process parameters up to 90% of particles non-spherical shape and different inner structure may be formed. Production of nanoparticles of desired shape, and structure

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can be another important step toward their more versatile use. However, the preparation of clusters of certain morphology is a formidable technical task. To solve this problem the method of heat treatment of obtainable metallic particles can be used as a final stage of synthesis from gaseous phase.

It is necessary to have precise information on morphological changes in particles under study throughout the heating process to create the most effective process of thermal action. For this purpose two methods of particles heat treatment studying are experimentally used—*differential electrical mobility analyzers* (DMA) and *transmission electron microscopy* (TEM) (Tsyganov et al., 2007).

These methods are appropriate for particles concentration, diameter and morphology determination before and after heat treatment, but they fail to provide total information on the processes which occur during materials heating and phenomena which affect structuring and shaping of the treated particles. Only computer modeling can provide such information to the full extent. Therefore, the paper provides the results of the modeling of heat treatment of nanosized copper particles synthesized from gaseous phase in order to produce particles of proper shape and internal structure.

One of the first theoretical attempts of the detailed description of the process of sintering of nanoparticles was realized by Koch and Friedlander (1990). Koch and Friedlander developed a self-preserving model for the sintering by assuming the minimization of the agglomerate surface as the essential driving force for the sintering process. Based upon these studies, models for the mutual interplay between interparticles coalescence and coagulation of agglomerates were developed (Kruis et al., 1993; Seto et al., 1995).

In several articles (Zhu & Averback, 1996; Hendy et al., 2003; Arcidiacono, Bieri, Poulikakos, & Grigoropoulos, 2004) the sintering process of single-crystal spherical nanoparticles of metals (Cu and Au) of different sizes in an inert-gas atmosphere and vacuum were simulated with the help of molecular-dynamics method. In other studies (Alarifi et al., 2013; Buesser & Pratsinis, 2015) sintering of composite Au nanoparticles, composed of two, three and four primary spherical clusters was studied. It was found, that the sintering process is realized in two stages-surface diffusion and plastic deformation nearer to the melting point.

However, in all the works presented above, the sintering processes were studied for nanoparticles, which consisted of single-crystal spherical clusters cut out from the perfect lattice arranged at short intervals from one another. In our work, while studying the processes of thermal effect, Cu nanoparticles synthesized from the gas phase and therefore having different sizes, shape and degree of agglomeration will be used for the first time.

2. Simulation technique

This section gives a description of the basic principles of the MD procedure used in this study. Any simulation method must first choose some potential of interaction between atoms in the system under consideration. The interatomic potentials used in computer simulations range from *ab-initio* to simple Lennard-Jones type empirical ones.

The choice of a particular potential depends on the problem formulation, the properties to be examined, the available computing resources, and the desired accuracy of results. Upon analysis of various representations of the potential energy of interatomic interaction, we decided to simulate the formation of Cu nanoclusters from the high-temperature gas phase in terms of the well-established modified tight-binding TB-SMA potentials as introduced by Cleri and Rosato (Cleri & Rosato, 1993). According to this model, the potential energy of the atomic system is calculated using the following expression:

$$E=\sum_i\left[-\left(\sum\nolimits_{i\neq j}\xi_{\alpha\beta}^2 e^{-2q_{\alpha\beta}\left(r_{ij}/r_{\alpha\beta}^0-1\right)}\right)^{1/2}+\frac{1}{2}\sum\nolimits_{i\neq j}A_{\alpha\beta}e^{-p_{\alpha\beta}\left(r_{ij}/r_{\alpha\beta}^0-1\right)}\right], \tag{1}$$

where r_{ij} is the distance between ith and jth atoms and subscripts α and β indicate various types of atoms. The quantities $\xi_{\alpha\beta}$, $p_{\alpha\beta}$, $A_{\alpha\beta}$, $q_{\alpha\beta}$, and $r_{\alpha\beta}^0$, which determine the parameters of system elements are as defined in (Cleri & Rosato, 1993). We believe that this potential of interaction ensures quite adequate description of the process of metal cluster formation on time scales characteristic of its nucleation, coagulation, and subsequent growth and takes into account the main features of interatomic interaction in copper particles.

A simple and efficient computer program is employed in this study to simulate the evolution of several thousands of atoms over a time interval of several nanoseconds. This is impossible to achieve with more realistic models, such as *ab-initio* methods, even on high-performance computers.

The starting point of the condensation process was a spatial configuration consisting of 85,000 Cu atoms, which were uniformly distributed in a region of space with a volume of $V=42{,}600\ \mathrm{nm}^3$ and periodic boundary conditions. The atomic velocities were set according to the Maxwell–Boltzmann distribution at an initial temperature of $T_i=1000$ K, which is quite typical of the gas phase synthesis processes (Backman, Jokiniemi, Auvinen, & Lehtinen, 2002). In the inert-gas–metal–vapor mixture used to produce Cu nanoclusters by gas-phase condensation, the inert gas is mainly required to cool the metal vapor and control its temperature. In our approach, external influences are reduced to heat removal from the simulated system. At the start of the simulation, a supersaturated metal vapor fills the condensation region.

The main idea of the proposed gas-phase condensation model is to minimize the rate of formation of primary nuclei from a supersaturated metal vapor. Otherwise, single atoms or small clusters will be captured on the collection surface. This

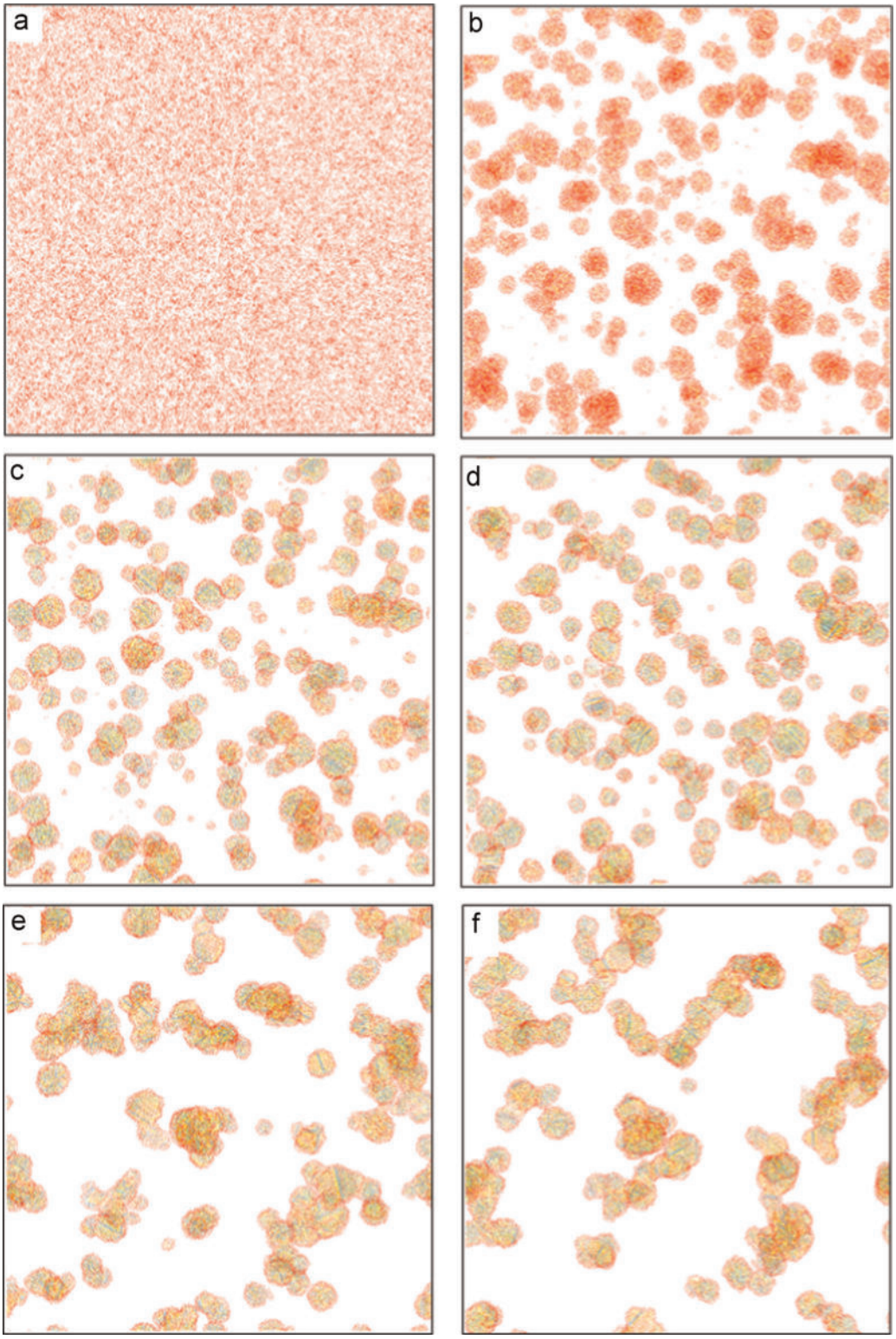


Fig. 1. Evolution of 85,000 Cu atoms from a hot atomic gas to condensed nanoparticles: (a) $t=5$ ps, $T=6939$ K; (b) $t=0.2$ ns, $T=1130$ K; (c) $t=0.4$ ns, $T=390$ K; (d) $t=2.2$ ns, $T=77$ K; (e) $t=10$ ns, $T=77$ K; (f) $t=20$ ns, $T=77$ K.

effect is avoided by using an inert gas in the condensation region to reduce the diffusion rate of atoms outside the supersaturated vapor source via an increase in collision frequency.

Another important issue in the simulation is related to the interaction of the atomic system with the heat reservoir. Since the formation of clusters from an atmosphere of individual atoms is accompanied by the release of considerable binding energy, this interaction is necessary to eliminate excess growth in temperature, which can reach 5000–6000 K at the initial stages of synthesis. In direct experiment, hot metal atoms are usually transported to the synthesis chamber with a flow of an inert gas cooled to the required working (synthesis) temperature. In the course of our simulation, the kinetic energy of Cu atoms was decreased using an Anderson thermostat (Pang, 2006) with control parameter W, in which the total system energy is calculated as

$$H=\frac{1}{2}\sum_{i=1}^{N} m_i\dot{\mathbf{r}}_i^2+E(\mathbf{r}_1,\mathbf{r}_2,...,\mathbf{r}_N)+\frac{W}{2}\dot{V}^2+P_0V. \tag{2}$$

According to this stochastic method, metal atoms exhibit random collisions with certain virtual particles so as to model the interaction with atoms of the cooling inert gas, the temperature of which under the adopted experimental conditions corresponded to liquid nitrogen temperature $T_f=77$ K.

In simulations of the heat treatment process of the metal clusters synthesized from the gas phase, the temperature was stabilized using a Nosé thermostat (Nosé, 1984). According to this, the total energy is a sum of the kinetic and potential energies of particles and the kinetic and potential energies of the heat reservoir s:

$$H=\frac{1}{2}\sum_{i=1}^{N} m_i\dot{\mathbf{r}}_i^2+E(\mathbf{r}_1,\mathbf{r}_2,...,\mathbf{r}_N)+\left(\frac{d\ln s}{dt}\right)^2\frac{Q}{2}+(3N+1)k_BT_0\ln s, \tag{3}$$

where Q is the control parameter. Using this approach it is possible to set the temperature accurate to within 1 K, which is very important, for example, in determining the melting temperature of a metal cluster (Gafner et al., 2012). The temperature was determined from the average kinetic energy of atoms, which was calculated using the high-speed Verlet algorithm (Pang, 2006) at a time step of $h=2$ fs. The processes of condensation and subsequent heat treatment were simulated using the MDNTP program developed by Dr. Ralf Meyer (University of Duisburg, Germany).

3. Results and discussion

The formation of a cluster in a metastable gas phase starts with homogeneous nucleation (Smirnov, 2010). After termination of this process, the subsequent growth of clusters can proceed according to various mechanisms. The most important of these are agglomeration and coalescence. In the case of agglomeration, clusters combine without changing their shape, whereas during coalescence, clusters fuse and merge together with the formation of a common particle with a shape differing from that of the initial clusters prior to their collision.

Our analysis of snapshots of the simulated system reveals four distinct stages of its evolution from a hot atomic gas to a few synthesized particles (Fig. 1 and Video 1). The release of binding energy during the initial stage of cluster nucleation raises the temperature to a maximum of $T=6939$ K. At this stage, the state of the system can be characterized as a hot gas of metal atoms, as illustrated in Fig. 1a.

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The use of the Andersen thermostat gradually reduces the system's temperature, and copper atoms begin to condense into hot liquid droplets. The temperature reached in the course of a simulation can be so high that fast-moving atoms cannot be bound to a cluster by attractive forces. Thus, one or several atoms may escape and move away from an $(N-1)$-atom nanoparticle with constant velocity. However, because of the periodic boundary conditions, these atoms cannot leave the simulation domain. Eventually, such an atom is attracted by, and joins, an already existing liquid nanoparticle. An analogous mechanism of aggregation is used in the case of liquid-phase nucleation.

At the time $t=0.2$ ns after the start of the simulation, when temperature drops to $T=1730$ K, almost all copper atoms have condensed into spherical liquid clusters, as in Fig. 1b. The largest cluster in this image contains 1855 atoms, 93% of all atoms are in clusters larger than 100 atoms, and 82% are in clusters larger than 200 atoms. Even though many clusters become denser, this stage can be interpreted as coalescence since liquid aggregating clusters immediately lose all memory of their previous states.

At a lower temperature, most primary particles aggregate into large nonspherical clusters, which is clear from Fig. 1c. Note that both free atoms and clusters smaller than 50 atoms completely disappear by $t=0.98$ ns. During this particular stage, nanoparticles crystallize into "frozen" clusters, even though some changes in cluster morphology are still possible at its very beginning.

Fig. 1d shows the system at $t=2.2$ ns, when temperature reaches its final value $T_f=77$ K. By this time, most small clusters disappear, and 97% of all atoms are in clusters larger than 200 atoms, while the largest cluster contains 2412 atoms. By the final stage ($t=20$ ns), most primary particles have aggregated into large clusters, as in Fig. 1f. The thirty five nanoparticles remaining at this stage contain 81–7218 atoms. It is seen that particles produced from gaseous medium have the most bizarre shape and wide range of segments of different crystal structure. For that matter the relevant issue is further

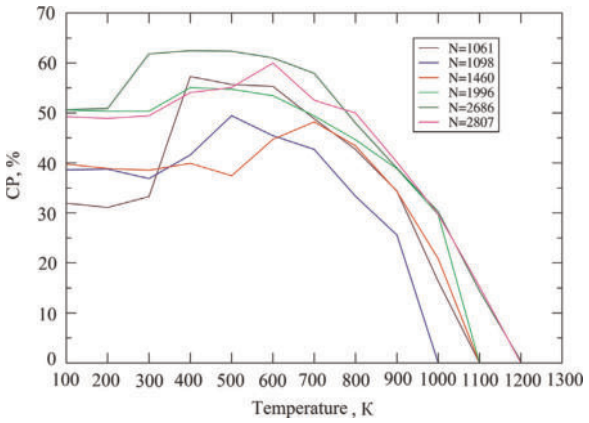


Fig. 2. The fraction of the number of atoms with coordination number $z=12$ (in %) of total N atoms in the cluster as a function of temperature for a cluster size of 1061–2807 atoms.

processing of synthesized nanoparticles in order to get fixed shape and structure. The molecular dynamics simulations of the heating processes of copper clusters using the Nosé thermostat was carried out to determine the main features of particles heat treatment.

To study particles shape and structural properties depending on temperature the gradual heating of copper nanoclusters available from condensation from 100 K to 1200 K with temperature steps of 100 K was modeled. For each temperature particles were held from 3 to 4 ns depending on the cluster size. Clusters under study were chosen in such a way that to fully reflect differences in size, original shape and structure of particles available from condensation. In this simulation, we revealed two typical tendencies:

1. If a particle has a regular internal structure, when heating a cluster to temperature 300–500 K further structure ordering takes place and the number of atoms with coordination number $z=12$ (CP) increases to 55–60%. Long-range order destruction begins at $T=600$ K, the complete particle melting occurs at $T=1100$ K (Fig. 2).
2. In the absence of the strongly marked closely-packed core of cluster due to intensive kinetic processes under heating, the formation of such core occurs in the temperature range from $T=300$ K to $T=700$ K. With further temperature rise long-range order destruction occurs with the process termination to 1100 K (Fig. 2).

Relatively low maximum values of atoms with coordination number $z=12$ in particles under study (not more than 65%) is attributed to the fact that surface atoms were disregarded. The atoms share in a cluster of ideal spherical form is determined from the formula $N_s/N=4\cdot N^{-1/3}$ (where N is the number of atoms in a particle, N_s is the number of surface atoms) and can reach approximately 40% for a cluster consisting of 1061 atoms and approximately 29% for a cluster, consisting of 2807 atoms. Taking it into consideration we can conclude that in condition of simulated thermal influence there is practically complete ordering of synthesized particles internal structure.

To explain such a behavior (Fig. 2) in details the processes of clusters atomic restructuring in condition of thermal influence is analyzed. In Fig. 3 there are “snapshots” of Cu_{2686} cluster evolution at short-term thermal influence. This cluster pertains to the first group of particles having explicit close-packed core. The colors of the atoms denote the result of a CNA analysis (Meyer et al., 2005) as follows: atoms drawn in blue (green) have a local crystalline fcc (hcp) environment; yellow atoms have no identified structure, however their coordination number is $z=12$. Atoms with coordination numbers $z\neq 12$ are drawn in red.

We can see that in the early stages (Fig. 3a) the particle consisted of three combined with each other clusters with various internal organization. However, at $T=300$ K (Fig. 3b) there are distinct processes of transforming internal structure of amorphous area dividing the biggest initial cluster and element with face-centered cubic and hexagonal close-packed structure.

As a result of heating up to $T=500$ K the processes of intensive thermo-activated self-diffusion of surface atoms in the places of connection of two cluster parts begin, and by $T=700$ K this leads to formation of unified oval exterior form for this particle (Fig. 3d). The further increasing of temperature just slightly changes the exterior form and in accordance with the principle of surface energy minimum the particle is aiming to take spherical shape. At $T=900$ K the melting processes begin, in consequence of which the destruction of long-range order with completion of process at $T=1200$ K occurs (Fig. 3f). This behavior was neither very complex nor unusual, since the driving force for the combination of small clusters and formation of a common structure of the initial agglomerated particle was the existence of a large disordered region involving almost the entire Cu_{2686} cluster.

As an example of particle with absence of strong close-packed core the cluster of copper of 1460 atoms was selected (Fig. 4 and Video 2). After the completion of condensation process this cluster had distinct chain (wormlike) shape often

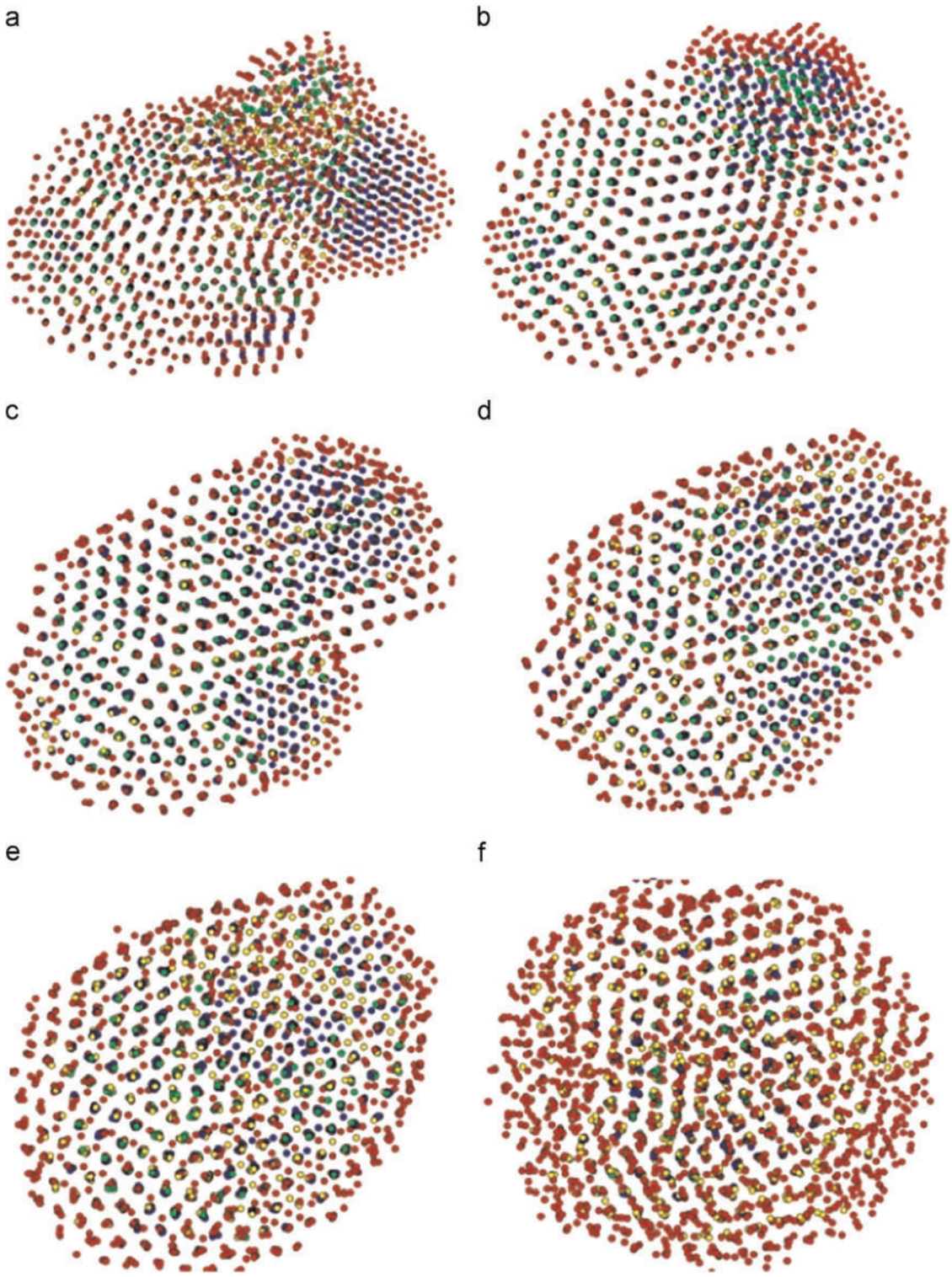


Fig. 3. Evolution of a Cu_{2686} cluster with increasing temperature: (a) $T=100$ K, (b) $T=300$ K, (c) $T=500$ K, (d) $T=700$ K, (e) $T=900$ K, and (f) $T=1100$ K. (For interpretation of the references to color in this figure, the reader is referred to the web version of this article.)

observed in real experiments on vapor phase synthesis of metal nanoparticles. Such an exterior shape is being a direct consequence of sintering processes taking place at low environment temperatures in nanoparticles condensation chambers. In our case the four initial clusters with various type of atoms crystalline ordering have agglomerated.

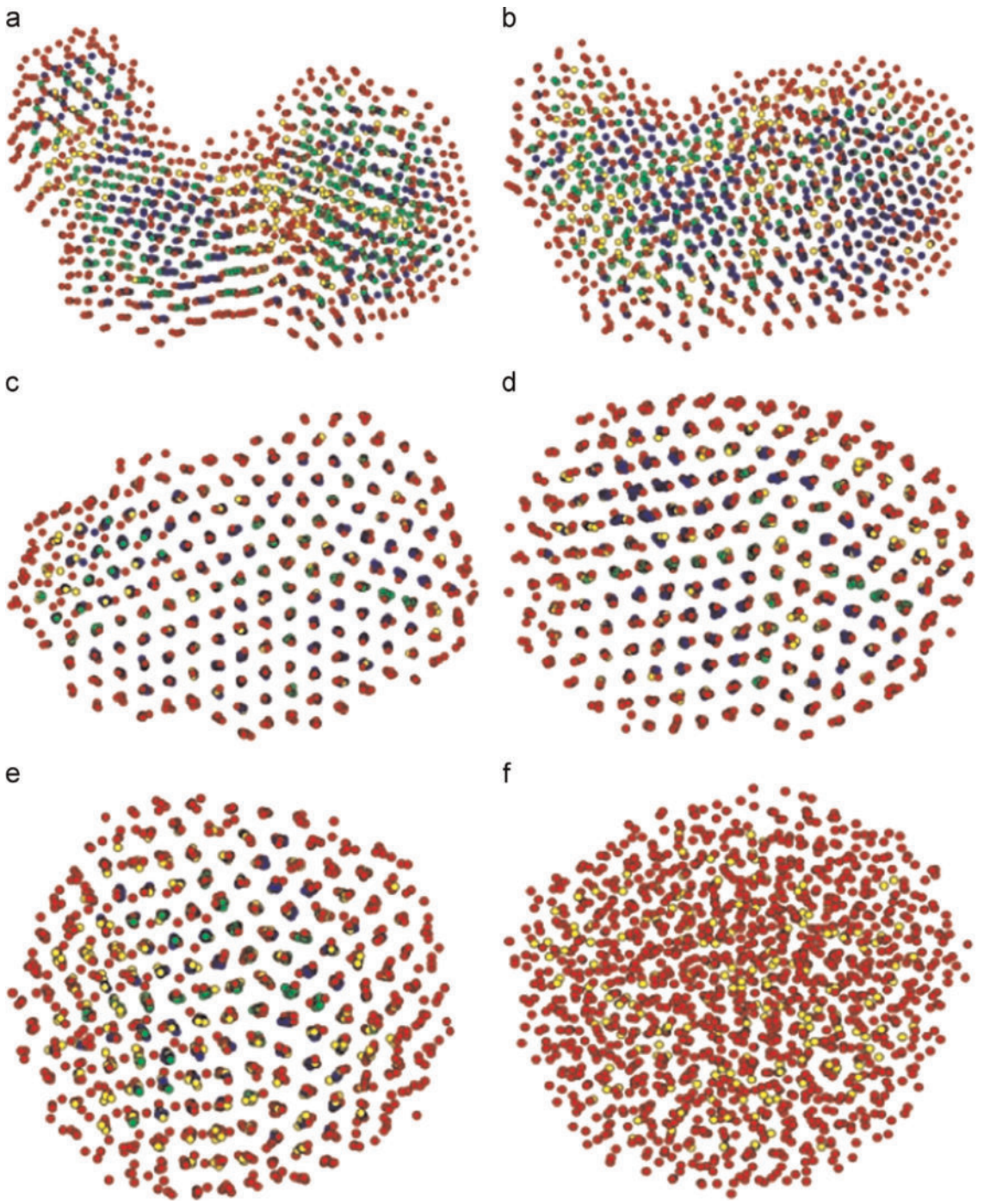


Fig. 4. Evolution of a Cu_{1460} cluster with increasing temperature: (a) $T=100$ K, (b) $T=500$ K, (c) $T=700$ K, (d) $T=900$ K, (e) $T=1000$ K, and (f) $T=1100$ K.

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Starting with $T=100$ K and up to $T=400$ there are almost no changes in cluster Cu_{1460}, however in the case of temperature increasing up to $T=500$ K its shape begins changing from chain to spherical. The changing of cluster structure is also represented with decrease of number of atoms with coordination number $z=12$ in the particle shown in Fig. 2. The intense processes of packaging caused by thermal treatment of the particle lead to insignificant destruction of internal structure of the layer connecting two cluster parts and, as a consequence – to decrease of close-packed atoms number (Fig. 4b). At $T=700$ K the cluster takes spherical shape with explicit common decahedral structure (Fig. 4c), which is conserved up to the melting temperature.

Note that the Cu_{1460} cluster significantly differs from Cu_{2686} in both external shape and internal structure, since the former consists of larger fragments belonging to significantly different structural types. However, in the main driving force for the combination of primary particles was again the existence of a rather large disordered (amorphous) region between the two largest primary clusters. The presence of this disordered spacer between agglomerated clusters facilitates the fitting of their crystal lattices. As a result, the thermodynamic competition is won by the structure (in this case, decahedral) that ensures the minimum energy. This is confirmed by the fact that even slight heating of the Cu_{1460} cluster (to $T=200$ K) leads to the rearrangement of this amorphous region, which eventually results in the transformation of the entire cluster. Such results of change of internal structure and external form of wormlike clusters are also observed in experimental research of staged thermal treatment of transition metals nanoparticles (Magnusson et al., 1999).

For all the six researched particles shown in Fig. 2 with temperature increasing the share of atoms with coordination number $z=12$ grows and peaks by the temperature from 400 to 700 K depending on particle's size and ordering of internal structure. Thus, to improve the internal morphology of particle by thermal influence method there is no need in increasing temperature over $T=700$ K, although the external configuration of cluster in this case does not always take perfect spherical shape and can remain at restructuring intermediate stage presented by ellipsoid.

In such a way several dozens of various clusters were thermally treated, and after analyzing the results we can conclude that the thermal processes influence the formation of the cluster external appearance and structure quite much. Although the observed particles of copper had different form and internal structure there was significant grading of such difference at higher temperatures.

However, not all investigated particles were influenced by the thermal process in cluster form and internal structure. The analysis has shown that thermal treatment in case of about 30% of clusters synthesized from gaseous environment had led neither to the formation of unified structure, nor to the formation of spherical shape. As one of the typical examples in Fig. 5 there is a Cu_{1771} cluster that during the whole process of heating constituted two particles with various internal structures divided by an amorphous layer. Being heated from $T=100$ K up to $T=800$ K the parts of the cluster have adopted almost spherical shape with simultaneous reduction of contacting area (Fig. 5c). The particle is fully dissolved without packaging of interacting segments at further increasing of temperature (Fig. 5e and f).

It was suggested that such a behavior of particles at thermal treatment can be the result of sintering process at rather low temperatures of perfect spherical clusters with icosahedral structures. Because of the low temperature of kinetic energies environment the collision of particles was insufficient for the activation of surface atoms interdiffusion, and the clusters were just "sticking" to each other due to their interacting forces. As the icosahedral structure corresponds to minimal value of energy, at further heating the initial lightly connected icosahedral clusters saved such a construction of atoms and simultaneously turned into maximal possible sphere. We should also note that clusters gained through direct experiments on condensation from gaseous phase often possess segments with icosahedral structure. In order to verify this assumption, we have studied other nonunified clusters. It was found that, in the overwhelming majority of cases, these clusters comprised combinations of several icosahedral clusters that agglomerated at very low temperatures (below 100 K). The competition of these structures in a cluster leads to further formation of an amorphous-like layer between the particle parts.

Thus, the results of computer simulations of the thermal treatment of copper clusters synthesized from the gas phase suggest that a rather high temperature of agglomeration is the key factor. If the agglomeration took place at the initial stages

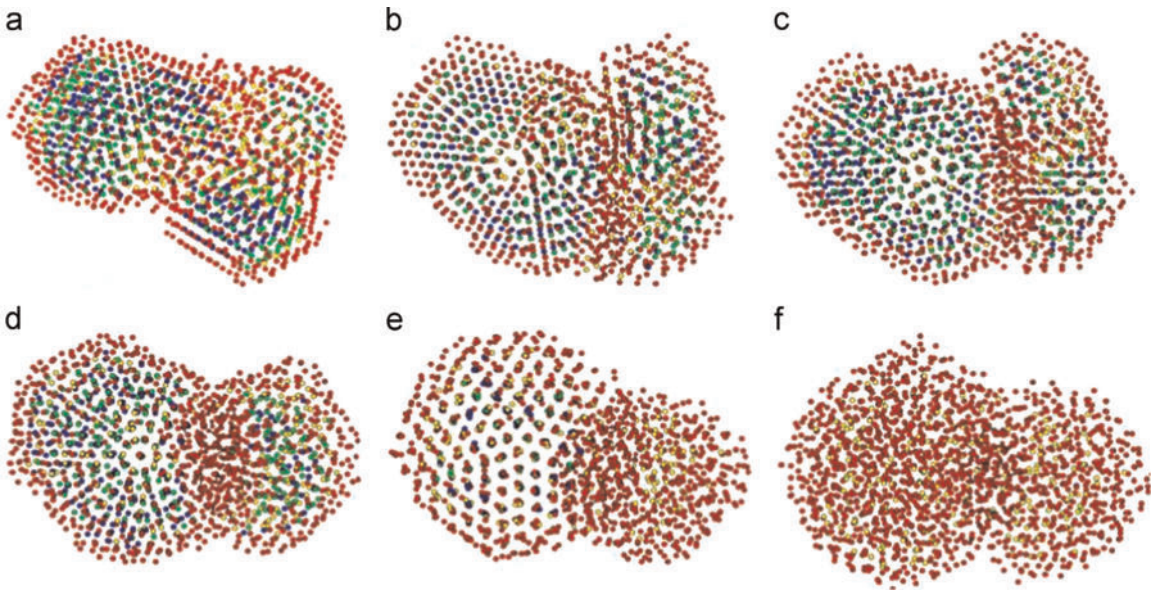


Fig. 5. Evolution of a Cu_{1771} cluster with increasing temperature: (a) $T=77$ K, (b) $T=100$ K, (c) $T=500$ K, (d) $T=800$ K, (e) $T=1000$ K, and (f) $T=1200$ K.

of development, a high ambient temperature favored the agglomeration of primary clusters into almost unified particles. If the agglomeration proceeded at low temperatures, the kinetic energy of collision was insufficient to fuse the primary particles together. As a result, the parts of the cluster merely stuck together, predominantly due to interatomic interaction forces. In this case, the initial shape of the cluster changed during its thermal treatment and acquired a chainlike or wormlike structure.

4. Conclusion

In this paper the processes of copper atoms condensation with formation of nanoparticles of various form and structure were modeled with the method of molecular dynamics with using of tight-binding modified potential TB-SMA. To define the mechanism of external form and internal structure evolution, the copper nanoclusters condensed from gaseous phase were influenced by thermal treatment.

As a result of thermal treatment modeling we have found that in case of 70% of observed clusters there was full ordering of internal structure and external form. The remaining 30% has failed to form unified shape and structure in the process of thermal-activated relaxation and was presented as competing parts of one particle with comparable energies of surface and chemical potentials. The analysis of the evolution of clusters synthesized from a gaseous medium suggests, that it is necessary to restrict the temperature of cooling of the working gas mixture to a level of 400–500 K to obtain a large fraction of spherical metal nanoparticles with a unified crystalline structure. Cooling the mixture by liquid nitrogen according to the commonly accepted practice leads to the formation of a considerable fraction of weakly bound (because of the small area of contact) icosahedral clusters. Upon subsequent heat treatment, these clusters form wormlike nanoparticles consisting of segments with different crystalline structures.

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